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Paving the way for Brazil's first national rare diseases registry: the RARAS data governance model

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Abstract

Background Digital governance in health research is crucial for ensuring data quality, accessibility, and interoperability, particularly in rare diseases. Establishing a National Rare Diseases Registry in Brazil faces structural and technological challenges. The Brazilian Rare Disease Network (RARAS) proposes an innovative digital governance model to standardize, manage, and enhance the use of these data, aligning with the FAIR Principles (Findable, Accessible, Interoperable, and Reusable) to optimize scientific collaboration and public health decision-making.

Methods This study employed a multi-tiered governance framework integrating technological, ethical, and operational perspectives. The model prioritizes data standardization, implementation of FAIR principles, and stakeholder engagement. A structured Data Management Plan was developed using the ARGOS platform, and metadata compliance was assessed through the F-UJI tool. The network leveraged linked data technologies, ontologies, and REDCap for secure electronic data capture and management. The RE-AIM framework (Reach, Effectiveness, Adoption, Implementation, and Maintenance) was applied to evaluate the model's impact.

Results The RARAS governance model successfully improved data integration across 45 rare disease centers, enhancing interoperability and data-sharing practices. FAIR assessment using F-UJI indicated moderate compliance (50% FAIR score), with strengths in semantic interoperability but gaps in metadata completeness and accessibility. Training initiatives across Brazilian regions reached 463 participants, promoting awareness of data quality. Challenges included heterogeneous data collection practices, limited digital literacy among stakeholders, and the need for automated metadata validation. The RE-AIM analysis highlighted substantial reach and adoption but identified areas for improvement in long-term sustainability.

Conclusion This study presents a scalable digital governance model for rare disease research in Brazil, improving data standardization and integration. The findings underscore the need for continuous FAIRification efforts, enhanced

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metadata structures, and federated data-sharing strategies. The model provides a foundation for developing a robust national registry, supporting data-driven public health policies, and fostering international collaboration in rare disease research. Future work should focus on automation, federated learning, and expanding FAIR compliance to maximize the long-term impact of digital governance in health research.

Keywords Rare disease, Digital health, Governance model, National registry, Research network

Introduction

The digital transformation in healthcare has significantly reshaped how research is conducted and how public health challenges are addressed. The growing adoption of digital solutions, such as electronic health records, mobile applications, and online data collection platforms, presents vast opportunities for health research and clinical care [1]. However, this shift also underscores the urgent need to establish effective governance models in digital health research. In this sense, the complexity and scope of data generated require integrated approaches to ensure quality, safety, ethics, and effectiveness in a study [2].

Poor data governance in the digital health era is not merely an administrative issue. It directly impacts patient outcomes, slows medical progress, and widens the gap between research and practice. Without clear governance structures, patient data may remain underutilized, and the potential of digital innovation may not be fully realized [3]. The initiative's success in transforming scientific discoveries into real-world applications depends on technology and trust. A well-structured digital governance model must foster process confidence, ensure data is handled transparently, and empower collaboration that leads to meaningful scientific breakthroughs (Budin-Ljosne & Harris, 2015) [4].

In this context, data governance has emerged as a cornerstone for digital health, particularly in managing rare diseases (RD). Research in RD faces well-known challenges, like fragmented data, low prevalence, and lack of standardized protocols, which pose persistent barriers to effective research and clinical decision-making. Previous studies have emphasized the importance of structured data governance in overcoming these limitations and enabling the creation of high-quality registries and interoperable datasets [5, 6].

Research networks have become essential for addressing these complexities in developing digital governance models. These organizations can elucidate the facilitating aspects or barriers to successful implementation [7]. Sharing high-quality data across research networks enables the large-scale evaluation of strategies and the global dissemination of equitable knowledge [8].

Collaborative initiatives, such as the European Reference Networks (ERNs), demonstrate how international efforts can help reduce the gaps between evidence and practice by promoting data quality and integration [9].

Such networks provide the infrastructure and shared databases needed to support multi-center studies and align research goals with the needs of both patients and experts.

The Brazilian Rare Disease Network (RARAS) was created to meet similar needs in Brazil. It combines reference centers for RD, university hospitals, and neonatal screening services across the national public healthcare system to establish a comprehensive national registry and foster collaborative studies. The network addresses key barriers, including inconsistent data collection, low digital literacy, and the absence of unified metadata standards [10].

RARAS also aims to implement a digital governance framework grounded in the FAIR (Findable, Accessible, Interoperable, and Reusable) principles to support interoperability and data reuse. These principles are key strategies for promoting high-quality, collaborative, and transparent scientific research, especially in interdisciplinary areas such as RD [11, 12].

Through research that optimizes health interventions and contributes to the alignment of more effective policies [13], RARAS provides the infrastructure and database to support diverse studies [12]. Considering data collection analysis and translating results into practical and effective actions for public health [14], the first Brazilian network ensures research priorities align with patient and expert concerns.

This study presents the RARAS digital governance model in the context of digital health, detailing its development, implementation, and evaluation. It aims to consolidate diverse practices into a harmonized framework that promotes ethical and practical use in advancing RD research in Brazil [12]. The solution targets local, regional, and national decision-makers responsible for managing and coordinating RD research and care in Brazil and health professionals seeking collaborative digital strategies to address data quality concerns related to reliable research and information on patient care and their families [10]. We aim to address the following research question: *'How can a digital health governance model support the long-term maintenance and availability of a national RD registry in Brazil while ensuring data quality, interoperability, and collaborative research?'*

Methods

Institutional and technical readiness assessment

This study employs a multi-tiered governance framework integrating technological, ethical, and operational components [5]. The framework includes data standardization protocols, stakeholder engagement strategies, and the application of FAIR principles, deploying advanced tools such as semantic web technologies and ontologies [15].

To ensure that the metadata and digital records adhere to these principles, we utilized the FAIRsharing platform [11]. This trusted repository provides persistent identifiers and structured metadata, facilitating data discoverability and accessibility [9]. It is also integrated into the RARAS portal to promote metadata quality and compliance. However, this study did not use the platform as a primary FAIR Data Point.

As part of the governance model's development and the foundation for a FAIR Data Point, we conducted two structured surveys to assess institutional digital maturity and technical readiness among RARAS stakeholders [16]. The first survey focused on evaluating infrastructure, digital workflows, and standard operating procedures [12]. The second survey targeted IT teams and coordination staff, assessing interoperability practices, metadata standards, and system configurations [16].

Both instruments were originally deployed in earlier phases of the RARAS project and were reapplied in 2023 and 2024 to monitor progress. Selected results from the initial rounds provide the basis for our longitudinal assessment and contextualize the evolution of the FAIR-aligned governance model [7]. Comparative analyses with the most recent data allowed us to identify key improvements and persistent gaps [8]. Together with the instruments, protocols, and digital assets described in the following sections, these findings were used to evaluate the model's impact. Full versions of the survey instruments and anonymized response data are available in a public repository.

RARAS stakeholders and digital assets

The RARAS health information ecosystem comprehensively integrates various digital components to support clinical and research purposes [10]. This ecosystem aims to provide patient-focused services while ensuring efficient data management and operational benefits for research. The main components of this ecosystem include four modules for participant registration and follow-up, analytical and visualization tools, data quality monitoring, and a human resource training platform [8].

The IT team members and project coordination staff fulfil the data steward's responsibilities. These include ensuring metadata quality, performing validation checks, and managing documentation [15]. This role involves

configuring Electronic Data Capture (EDC) system validations [17], maintaining the Data Management Plan (DMP), and overseeing institutional training and procedures. The team also manages workflows using tools like Trello and SCRUM, enabling structured task assignment, follow-up, and sprint reviews. These elements collectively ensure that stewardship functions are fully integrated into the project's operational structure [12].

The RARAS network aggregates and manages both retrospective and prospective data. Specifically, it includes anonymized retrospective information from 12,530 participants (2018–2019) and prospective data from 7,745 patients (2022–2024). These datasets were obtained from publicly available and institutionally shared sources, anonymized before analysis, and used under ethical guidelines for non-identifiable health data [18]. Therefore, individual patient consent was not required, as the study used only anonymized and non-identifiable data, in accordance with ethical guidelines [10].

DMP and EDC infrastructure

At the system's core is the integration of linked data and heterogeneous databases, which enables the connection and integration of multiple data sources. This integration is essential for maintaining a unified and coherent data structure that supports various clinical and research activities. We developed a DMP using the ARGOS platform to ensure structured and well-documented data management. The ARGOS tool provided a structured workflow for defining data collection standards, storage policies, and access controls, as well as facilitating long-term data governance [15]. Although they have limited data-sharing capabilities [11], the use of electronic systems for data collection and analysis has been growing over the last decade.

EDC systems mitigate risks associated with paper instruments, enhance the collection of high-quality data necessary for health research, and reduce the cost and time required for monitoring and data management activities. To build and manage online surveys and databases that capture research and administrative data within the network, we used the Research Electronic Data Capture (REDCap) system, a free, stable, widely used, and well-documented software offering Application Programming Interfaces (API) for integration with other systems [17].

Although enabling data governance, tools like REDCap and FAIRsharing do not guarantee it on their own [19]. Their effectiveness depends on appropriate access controls, quality checks, and configurations defined by the registry coordination team [15]. Metadata validation is carried out through internal quality control procedures during data collection and curation [6]. The FAIRified data in the RARAS platform is semantically queryable

via a SPARQL endpoint, which generates a virtual RDF graph over a relational database and uses a custom ontology to enable semantic search. Additionally, structured metadata embedded in HTML using RDFa and Microdata can be extracted with tools such as Apache Any23. Further details are presented in the FAIRification process section.

Instruments and monitoring process

The data collection instruments were established in accordance with the National Policy for Comprehensive Care for People with Rare Diseases [20] within the Brazilian public health system context [20] and the WHO guidelines and recommendations for digital health interventions [2]. In addition, the instruments were clinically validated by experts from the RARAS network based on a systematic review to verify all minimum data sets (MDS) for RD used in healthcare networks worldwide. MDS is a public health strategy adopted by many countries to standardize essential data for their health systems [21].

The review included 20 studies that addressed MDS structures for RD and organized data elements into 10 categories aligned with the WHO digital health guidelines [2]. The results guided the selection and categorization of variables most relevant to the Brazilian RD context and public health system. Then, our MDS includes demographic, clinical, and genetic pertinent variables to RD surveillance and care. We define assessment criteria based on a conceptual model for each MDS category [21], normalized according to FAIR principles.

The centralized data management and monitoring process adopted in this study involves continuous remote evaluation of collected data, conducted periodically by a qualified team designated by the study coordinator [18]. Following international best practices in human research, defined roles include a data manager, coordinators, monitors, local focal points (where applicable), and data entry personnel, often comprising academics and health professionals [22]. REDCap ensures confidentiality and compliance through configurable access controls and a Data Resolution Workflow module, which implements data quality rules, facilitates query creation and response, and resolves data quality issues [17].

REDCap facilitates automated checks across the entire database to identify inconsistencies and generate reports, allowing users to open and assign queries from these reports [19]. Patient numbers and unique identifiers are used to prevent duplicate records, especially in multicenter studies. This ensures a single participant is not counted multiple times during data analysis, avoiding potential biases. Verification strategies may combine composite identifiers, such as the participant's name, mother's name, and date of birth, to detect duplicities [14].

The ecosystem centralizes and disseminates the integrated and public information through a web portal, serving as a hub for data sharing and communication among stakeholders. Thanks to an accessible and reliable web portal and the possibility of broad dissemination, it aims to meet the needs of an audience with potentially low digital literacy, specifically patients and their family members. This ensures greater engagement with more accurate results, optimized, and available in various formats (e.g., text, audio, and images) [23, 32].

The RARAS portal stores various data types and metadata on research studies. Currently, the portal does not collect Patient-Reported Outcome Measures (PROMs). The RARAS portal focuses on patient demographic information, clinical outcomes, and epidemiological data related to RD from reference centers. However, plans are underway to incorporate PROMs in future platform iterations to enhance patient-reported data collection.

At RARAS, implementing FAIR principles for research data is a multidisciplinary, collaborative effort to address the challenges associated with research and optimize data management and analysis. This unprecedented initiative in Brazil, launched in 2020, focuses on the FAIRification of priority resources and developing a FAIR-based architecture that complements existing data management systems at RARAS [6]. This effort also incorporates the RE-AIM evaluation framework to assess its impact and effectiveness. This framework ensures that the system meets its goals in real-world applications, providing valuable feedback for continuous improvement [23].

The fairification process

The FAIRification process is characterized by steps that ensure that scientific data are FAIR. Advancing scientific knowledge efficiently depends on ensuring that data is transparent and reproducible [24]. Despite its primary objective of addressing data, the FAIR approach can be extended to other resources, such as portals, services, and scientific tools, since data drives them. In scientific governance data models, the FAIRification process can be utilized to ensure that digital assets are structured in a manner that enables the rapid FAIRification of scientific information and resources [25]. Regarding the four principles of FAIR, we defined a process to ensure each one:

- **Findable:** To make data findable, descriptive metadata must be created, providing information about the content of the data, such as title, author, date, keywords, context, and other relevant details. Metadata makes data searchable and helps users locate it in repositories or databases. Using consistent and standardized descriptive metadata allows us to add structured data markup in JSON-LD format to portal application web pages [26].

- **Accessible:** We defined the FAIRsharing platform as a trusted repository with licensed use of and access to data and metadata. Access to metadata also plays a crucial role in ensuring data accessibility. They may include information about accessing the data, such as URLs, persistent unique identifiers (e.g., Digital Object Identifier - DOI), license information, and terms of use. This metadata ensures that other researchers can obtain and use the data [25]. We also utilize the LattesData platform for storing research data, a Brazilian open-access repository for scientific research data, which focuses on preserving and sharing data from CNPq-funded projects to support Open Science and enhance research reproducibility and efficiency. The established standard communication protocols include the Hypertext Transfer Protocol (HTTP) and Application Programming Interfaces (APIs) for applications, with the integration of Digital Object Identifiers (DOIs) adopted [27].
- **Interoperability:** Interoperability relies on the mutual understanding of data between different systems. To this end, metadata may include information about the data structure, file formats, controlled vocabularies, and ontologies [26]. This structured metadata helps ensure that data can be understood and integrated across different contexts. We adopt open standards, such as the Resource Description Framework (RDF) and the Web Ontology Language (OWL), and utilize secure HTTP (HTTPS) data transfer protocols for sharing methods [25]. A specific Internationalized Resource Identifier (IRI) was defined for the ontology in its header using Protégé, a widely used ontology editor and knowledge management tool.
- **Reuse:** Whenever possible, we adopted standardized terminologies and ontologies available for RD, such as Orphanet and the Human Phenotype Ontology (HPO) [28]. We also provide a data dictionary that includes controlled vocabulary and structured metadata in data formatting standards such as Comma-Separated Values (CSV) and JavaScript Object Notation (JSON). Furthermore, we provide a General Public License (GPL) that describes the rights and restrictions for using and reusing data, protects copyright, and promotes data reuse. The description contains the DMP and data analysis plans (DAP). It also includes details on how the data was generated and processed, providing information about the data gathering method, data processing, analysis, visualization codes, and intellectual property protection [10].

The semantic representation adopted in the RARAS platform was built based on the MDS defined earlier in this study, which was established following a systematic review and expert validation process [21]. We utilized a combination of ICD-10, Orphanet (Orphacode), and OMIM codes for clinical diagnosis. ICD-10 is the official classification system adopted by Brazil's Unified Health System (SUS), particularly for hospital admissions, epidemiological notifications, and administrative processing [17, 20].

However, since ICD-10 lacks the granularity required for many RD, a mapping process was implemented to align ICD-10 codes with Orphacodes, enabling broader interoperability with international RD registries [25]. OMIM identifiers were used to complement genetic and molecular-level descriptions. The HPO was also used to structure phenotypic descriptors [28].

For sociodemographic and socioeconomic data, we utilized standard variables defined by the Brazilian Institute of Geography and Statistics (IBGE), including region, race/color, schooling, and household composition [12]. We applied the 2024 ABEP Classification Index to assess socioeconomic status, which ranks households using a multidimensional score across income, education, housing quality, and access to services. This ensures standardized stratification of participants' living conditions, supporting equity analysis in RD care [21].

To enhance semantic clarity and machine readability, we applied widely adopted ontologies and metadata standards, such as Schema.org, Dublin Core Terms, Data Catalog Vocabulary (DCAT), and DataCite, which support findability and accessibility across platforms [25]. All semantic components were validated through an expert review process involving clinical professionals and technical leads from the RARAS coordination centers to ensure their domain adequacy and consistency with the Brazilian context and FAIR principles [26].

Evaluation approach

RE-AIM framework and FAIR activities

Even when evidence exists, the challenges related to implementing a new technology significantly influence how an approach proven under research conditions performs under real-world conditions. Implementation research (IR) provides evidence on the most effective ways to support the adoption and optimize the use of innovations. IR represents an essential interface between the availability of tools, strategies, and interventions and their use within a health system [29].

We applied the RE-AIM framework to evaluate the impact of the digital governance model. RE-AIM is a tool for exploring the impact of digital technologies, measuring their effect on five essential dimensions for successful implementation: Reach, Effectiveness, Adoption,

Implementation, and Maintenance [23]. Each dimension was assessed using predefined indicators derived from metadata quality, platform usage statistics, training reach, and adoption rates. The RE-AIM model provided a structured lens to interpret implementation and sustainability outcomes.

Reach refers to the number or proportion of individuals affected by the intervention. Efficacy refers to the outcomes of the intervention, including both positive and negative effects, as well as its impact on quality of life and other relevant aspects. It encompasses the reasons for the intervention's success or lack of positive results. Adoption refers to the proportion and representativeness of providers who adopt the intervention and the agents responsible for implementing it. Implementation refers to the elements of the intervention, including delivery on time and as intended, the adaptations made, and their acceptance by end-users. Maintenance measures the level at which an intervention is sustained over time, including the system, provider, and individual levels. These domains are essential to the ultimate impact of the new intervention, and each requires a different evaluation approach [30].

FDO concept and governance alignment

The FAIR Digital Object (FDO) concept expands the FAIR principles by enabling the creation, management, and reuse of digital resources that are uniquely identified, described with rich metadata, and integrated into interoperable platforms [31]. FDOs are at the core of the governance model implemented in this study, which structures data activities and the FAIRfication process into digital components published via the RARAS portal [25].

The model also reflects principles aligned with international efforts to make RD data reusable and accessible, as highlighted by the ERNs and related FAIRification strategies. These initiatives highlight the importance of semantic standardization, metadata quality, and technical capacity-building in managing fragmented and sensitive data within the RD domain. In particular, Tumiene et al. (2022) [32] emphasize the importance of consistent education and infrastructure development to support FAIR-aligned digital health practices across diverse systems and regions.

Engaging RE-AIM in data management requires the use of appropriate governance models. This study's defined model links the core RARAS activities and FAIR principles to the RE-AIM steps [33]. This model reflects and assimilates nine essential FAIR activities described by Kodra (2018) [6]. These activities employ various techniques, processes, and tools outlined in the literature, which are utilized in data quality acquisition, monitoring, and analysis [34].

F-UJI evaluation

The F-UJI FAIR Data Assessment Tool was used to evaluate the FAIRness of the digital objects in the RARAS portal. This automated tool assesses metadata based on key FAIR principles by leveraging machine-readable services, persistent identifiers, and established metadata standards such as DataCite and Dublin Core [27].

F-UJI was chosen for its ability to generate structured, objective, and reproducible evaluations, offering clear diagnostic feedback and actionable suggestions for improvement [26]. It supports the iterative nature of FAIR implementation, rather than assuming binary compliance. Its integration into the FAIRfication workflow provided a standardized approach for maturity assessment aligned with the indicators proposed by the GO FAIR initiative.

Results

FAIR status of patient data and challenges in existing RD centers

In Brazil, clinical data from RD services are often recorded using local codes that vary across hospitals and employ different terminological and synonym languages, as data collection in a clinical setting is primarily focused on care, rather than research [21]. Despite the lack of internal integration in RD services, mapping local codes to standard terminologies, such as ORPHA codes, HPO, ICD-10, and OMIM, is an unavoidable task that must be accomplished to achieve interoperability. Recent analyses have highlighted significant variability in data-sharing practices among RD service centers. The survey conducted in 2020 [10] and repeated in 2023 revealed that most surveyed centers initially reported that they did not share data internally ($N=27/41, 65.9\%$). However, a subsequent assessment revealed that eight centers (19.5%) had transitioned to a data-sharing approach.

The original medical record data used to produce health statistics and indicators were well structured ($N=36/41, 85.3\%$), but must be machine-readable, uniform, and global. Our strategy focused on promoting a data quality culture, as the standards are not usually implemented at the source, i.e., in the health service's IT systems. We registered the RARAS portal on the FAIR-sharing platform to address this challenge. This ensured that metadata and data resources within the network adhered to the FAIR principles. The repository is a reference point for aligning the digital health governance framework with internationally recognized standards [11].

During the analyzed project period (2020–2024), 48 virtual training sessions were conducted, engaging 463 participants across the five Brazilian regions, with an average attendance of 80 participants per session. These monthly interactions were fundamental to identifying

and addressing data quality issues and challenges in implementing FAIR principles. They also underscored the need for standardized practices and enhanced collaborative efforts to augment data quality in Brazilian

Table 1 Summary of technical and training assessments in RD management across institutions

Evaluation Metric	Statistical Measures	Key Results
Tools Usability	N=25 Mean: 3.0 (good) Median: 3.0 STD: 1.12	The tools' usability was evaluated as generally acceptable, with an average score equivalent to 60% of the maximum on a five-level ordinal scale ("good" to "excellent") and a median confirming this result. However, the standard deviation of 1.12 suggests notable variability in the user experience, indicating inconsistencies in tool performance or varying levels of familiarity among users. This variability underscores the need for further investigation into specific usability factors, such as training adequacy and tool adaptability, to ensure a more consistent user experience across the network.
Infra-structure Efficiency	N=25 Mean: 0.52 Median: 1.0 STD: 0.50	The evaluation of technological infrastructure shows assessments are conducted in 52% of cases, corresponding to a mean score of 0.52. The median score of 1.0 confirms consistency, while a standard deviation of 0.51 reveals slight variability in the consistency of infrastructure evaluations across centers.
Data Sharing	N=25.0 Mean: 0.24 Median: 0.0 STD: 0.44	External data sharing occurs in 24% of cases (mean=0.24). The median score of 0.0 shows that most institutions either consistently share or do not share data, with minimal variability, as reflected by a standard deviation of 0.44.
Technical Support	N=24.0 Mean: 0.29 Median: 0.0 STD: 0.46	Technical support is available in 29% of cases (mean=0.29). The median score of 0.0 indicates that most institutions consistently lack or provide support. This lack of support may negatively impact the implementation of best practices, particularly in ensuring data quality and compliance with good data practices like the FAIR principles [6]. The standard deviation of 0.46 reflects low variability in these practices.
Training on Tools and Data	N=25.0 Mean: 0.68 Median: 1.0 STD: 0.47	Training on tools and data is conducted in 68% of cases (mean=0.68). The median score of 1.0 indicates that training is either consistently provided or not across institutions, and a standard deviation of 0.48 shows low variability in the implementation of training programs.
Specific Training on RD	N=17.0 Mean: 0.24 Median: 0.0 STD: 0.44	Specific training on RD is offered in 24% of cases (mean=0.24). The median score of 0.0 indicates that most institutions either consistently provide or do not provide such training, with low variability, as shown by a standard deviation of 0.44.

RD research. While the coordination project group maintained three-hour monthly video calls for general updates, insights, and feedback discussions, specific video calls were also set up with the required experts, each tailored to the topic. These regular and iterative meetings with all stakeholders were necessary to enable the development of an optimal dataset, a high-quality model, and computational standardization. The goal is for awareness regarding FAIR and data quality to develop progressively in the community.

Over time, longitudinal monitoring of data completeness, standardization, and interoperability indicators was established through automated dashboards, periodic audits, and algorithmic detection of inconsistencies, enabling a structured evaluation of improvements [34]. One of the key findings was the lack of standardization in variable naming across studies, which led to data integration challenges and frequent issues, including duplicate records, incorrect or incomplete form completion, and missing patient identifiers [8].

To mitigate these challenges, an iterative computational standardization process was implemented, involving automated validation mechanisms in REDCap [35], ontology-based disease classification, and data-matching algorithms to detect and correct duplications across centers [36]. Additionally, a panel for duplicate record detection was created, allowing real-time cross-checking of patients' entries while maintaining data privacy compliance.

The governance model also incorporated structured feedback loops: each participating center received periodic reports highlighting errors, inconsistencies, and recommendations for standardization [35]. Further capacity-building initiatives included tailored training sessions for research teams facing persistent challenges, reinforcement of FAIR principles application, and targeted interventions to optimize data collection and processing workflows, such as implementing an e-learning module. These efforts significantly enhanced data integrity, interoperability, and reusability, paving the way for establishing a higher-quality national RD registry.

We registered the DMP for all these processes using the ARGOS platform to ensure structured and reproducible data management practices [15]. The DMP defines key aspects of data handling, including storage, access control, and ethical considerations [12]. By implementing this structured plan, we enhanced transparency and compliance with best practices in digital health governance, facilitating a more sustainable and standardized research infrastructure [28].

There is a common lack of knowledge about linked data technologies in the clinical domain. Our technical survey, presented in Table 1, demonstrates that additional training is necessary to understand the requirements for

enriching data and utilizing practices and tools to their fullest extent.

Tools Usability” was assessed using a five-level ordinal scale (“poor”, “fair”, “good”, “very good”, “excellent”). All other metrics were binary-coded (0 = No, 1 = Yes) and are reported as proportions. Percentages are used in the narrative text for readability.

The difference in the number of observations (N) across categories arises from varying response completeness and relevance levels among the surveyed institutions, reflecting differences in their engagement with specific aspects of data sharing, technical support, and training initiatives. To help data providers and users gain experience with these technologies, we provided comprehensive data and metadata documentation, tools to enrich their data and awareness, and their related applications in general and in the context of RARAS. One key challenge the network faced was implementing a FAIR research data management plan. This comprehensive plan involves data standardization, annotation, and enrichment through unique identifiers, controlled vocabularies, and thorough metadata descriptions, as represented by the FAIRification process.

Coordinated fairification

The FAIRification process, implemented in collaboration with RARAS colleagues, followed a structured,

multi-phase workflow to enhance the machine’s readability, global interoperability, and findability of the RARAS portal [9], as shown in Fig. 1. This process was divided into three stages: Pre-FAIRification, FAIRification, and Post-FAIRification [31].

In the Pre-FAIRification stage (Step 1), we defined FAIRification objectives, mapped stakeholders and available digital assets, and developed governance instruments, including the DMP and DAP. This preparatory phase also included data profiling (Step 2) and metadata analysis (Step 3), carried out in collaboration with data collectors, data scientists, and IT teams [9, 15].

The FAIRification stage (Steps 2a–5a) focused on improving data and metadata quality. In Step 2a, clinical datasets were annotated with ontology terms using real-time outlier detection, centralized monitoring mechanisms, and mappings to controlled vocabularies such as ICD-10, ORPHA codes, HPO, and OMIM [12]. In parallel, Step 3b applied tools like BioPortal to enhance metadata consistency. Steps 4a and 4b involved the development of an ontological model to link datasets semantically, using standards such as DCAT and Schema.org [26]. In Step 5a, semantic web technologies were employed to make metadata machine-readable and enable data linkage [26]. These activities were conducted iteratively, supported by validation tools and expert review.

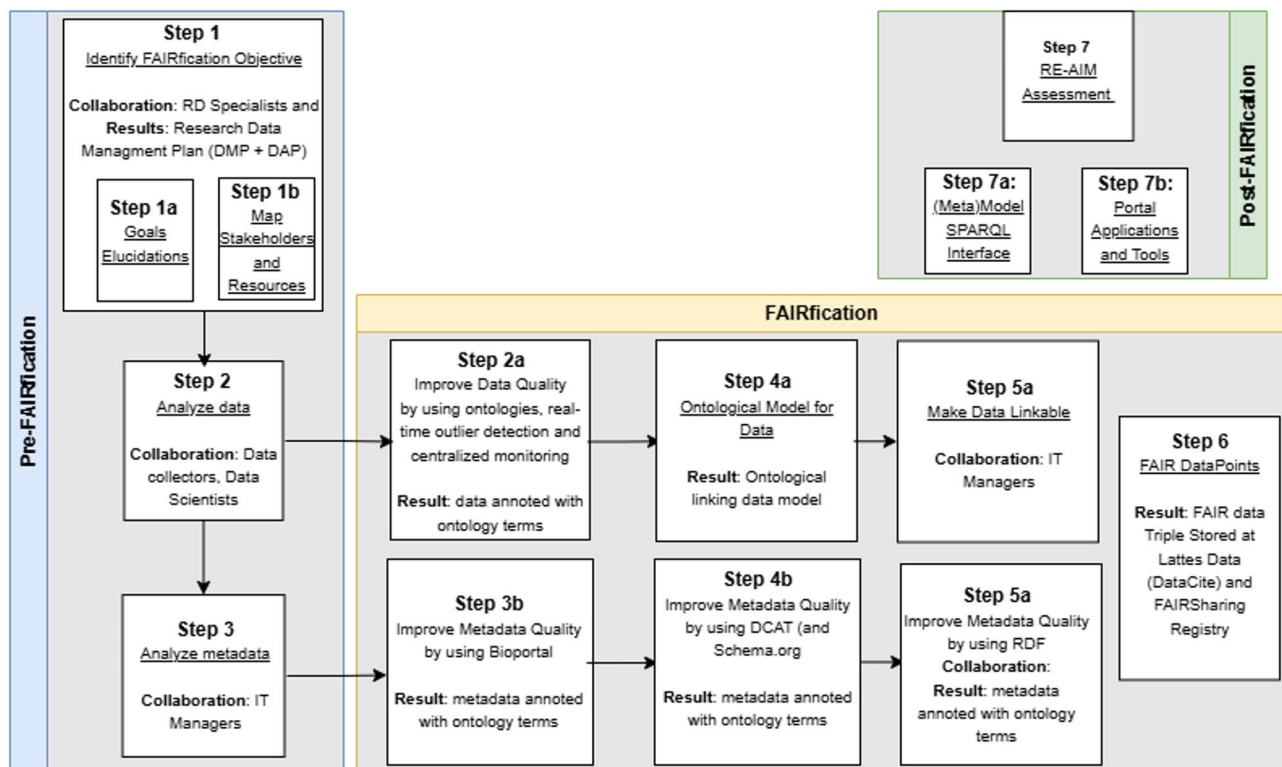


Fig. 1 RARAS FAIRification workflow to enhance data management and infrastructure in RD research

In Step 6, we registered curated metadata and datasets in trusted repositories (e.g., LattesData and FAIRsharing), assigning persistent identifiers to enhance findability and global interoperability [9, 27]. The Post-FAIRification stage (Step 7) included two components: (7a) assessment using the RE-AIM framework to evaluate the reach, adoption, and implementation of FAIR practices across centers, and (7b) the deployment of technical tools, such as SPARQL interfaces and the RARAS data portal, for ongoing access and monitoring [31]. These elements supported the continuous refinement of data governance policies and promoted reuse.

Semantic technologies and ontology development

As part of this FAIRification strategy, semantic technologies were implemented to enhance machine readability, global interoperability, and reusability of RARAS data. A coordinated effort involving data collectors, analysts, IT managers, and medical experts resulted in the development of domain-specific ontological models, enabling the structured representation of knowledge and semantic reasoning [26].

Using RDF and OWL, we created a formal structure to describe data entities and relationships within the RD domain. The resulting ontology, *portal_raras.owl*, is accessible through the IRI https://vocab.raras.org.br/portal_raras.owl, providing a persistent and machine-readable semantic structure. These technologies allow the data to be understood by humans and computational systems, supporting automated data integration and knowledge discovery [26].

To operationalize this, we adopted the D2RQ platform to map relational databases to RDF. D2RQ automatically generates ontology classes and properties from existing database schemas, enabling data export into a static RDF format. This approach ensures the structured and scalable transformation into FAIR-compliant resources [26, 31].

Furthermore, standard metadata vocabularies, such as DCAT and Schema.org, contributed to metadata enrichment and supported semantic interoperability, allowing for consistent interpretation and integration across different systems and institutions [26]. SPARQL interfaces enable federated queries on the portal's semantic layer, improving accessibility for researchers and stakeholders [26]. Integrating these semantic technologies into the RARAS portal architecture significantly advanced the platform's FAIR maturity, contributing to a sustainable and reusable national RD data infrastructure [9, 31].

RARAS digital governance model components

The central point in implementing FAIR principles within the context of RARAS was developing and adapting ontological models based on commonly used ontologies [28].

These models serve as a reference for the data stored in the source systems, effectively creating a comprehensive virtual data warehouse [33]. These models and the FAIR infrastructure were established to represent patient data as FDOs for analytical purposes, following the defined DMP [15] and DAP [31].

The establishment of the FAIR plan was achieved through a collaborative interdisciplinary approach. Our experience demonstrated that successful FAIRification necessitated the coordinated involvement of various key stakeholders, including data producers, analysts, consumers, and FAIR data modelers [25]. Their collective expertise was crucial in capturing the semantic meaning of the data through ontologies, highlighting the importance of their combined contributions in this process.

The experts manually preprocessed and validated the data and transferred it to REDCap [17]. Data modification was strictly controlled, requiring validation and formal authorization for record unlocking. The manager could only delete data upon proper justification. The system incorporates security and safety mechanisms to protect sensitive patient information collected during the research. These measures include data obfuscation, safeguarding against the inadvertent reidentification of patients, upholding data integrity, and ensuring that data privacy is upheld and all information is securely stored and accessed.

A robust monitoring and data validation infrastructure supported the high quality of the data and the expert's awareness of the importance of improving data generation and information sharing, facilitating managerial tasks. This infrastructure encompassed the REDCap Data Quality Module, a comprehensive semantic framework for data collection and management, and a computational pipeline for analyzing the produced data and detecting outliers [17]. The validated data were electronically and uniformly ontologically represented and added to the data collection forms to create a FAIR dataset [26].

Analytical and visualization tools are integrated to support data monitoring, analysis, and interpretation. These tools offer valuable insights into patient data and research findings, facilitating the development of effective strategies and interventions. Validation and control processes, using automated and manual checks, aim to ensure a high-quality, accurate, consistent, and applicable dataset throughout the study [33]. Articulating the RE-AIM dimensions presents an opportunity to identify synergies between the FAIR Principles and their framework, encompassing actions and responsibilities throughout the entire lifecycle of data and ecosystems [29].

Figure 2 presents an integrated representation of the RARAS data governance model, articulating the interplay between FAIR principles, the RE-AIM implementation framework, and the operational components of the

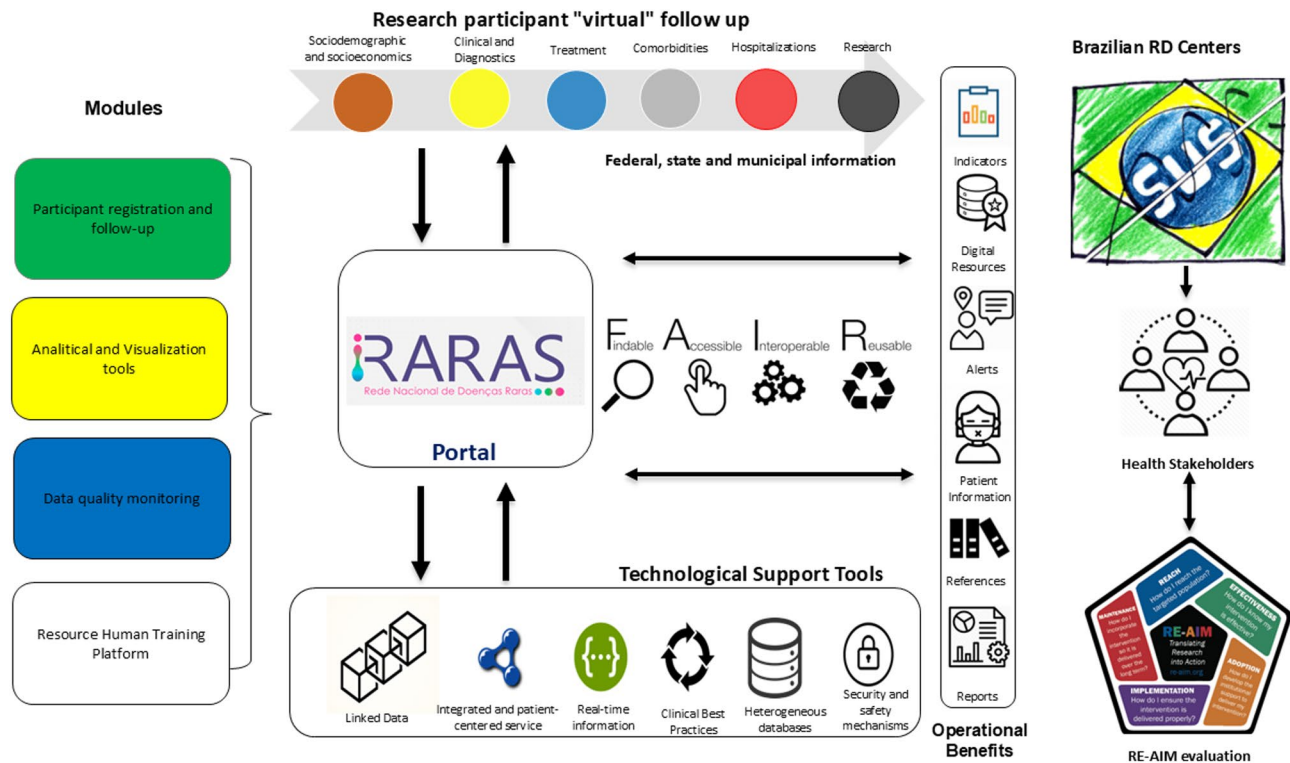


Fig. 2 Integration of FAIR Principles, RE-AIM Framework, and Governance model components

national RD platform. The RARAS portal is the central hub for data collection, integration, and dissemination at the model's core. It incorporates three key modules—participant registration and follow-up, analytical and visualization tools, and data quality monitoring—supported by technological tools, including linked data infrastructures, real-time information dashboards, clinical best practices repositories, and mechanisms to ensure data security and privacy.

The outer layer of the model anchors the governance approach in the FAIR principles, which inform the design and implementation of each module. The right-hand side of the diagram depicts the bidirectional data flow between the portal, national RD centers, and health stakeholders. The bottom layer embeds the RE-AIM framework as an evaluative structure to monitor the platform's performance, adoption, and sustainability. This multidimensional model illustrates how technological infrastructure, semantic data standards, and implementation science can be effectively integrated to inform FAIR-aligned governance systems in RD research and care.

RARAS FDO assessment and model evaluation

Our FAIR assessment of the data from the 45 centers revealed that while the structure and findability principles improved at each stage of the data management lifecycle in the local centers, no FAIR standards were applied to make the raw data and metadata globally

understandable 'to machines,' such as automated processing computers. As a result of the developed data architecture, the RARAS Dataset, combined with the RARAS RDF Schema and data following this knowledge representation, fulfills most FAIR principles (summarized in Table 2).

To evaluate RARAS's FDO implementation, we conducted an automated assessment using the F-UJI FAIR Data Assessment Tool. The evaluation provided insights into how RARAS adheres to the FAIR principles, identifying strengths and areas that require further attention. These results informed refinements to the governance framework and supported the ongoing improvement of the RARAS platform's FAIR capabilities. The assessment assigned RARAS FDO a total FAIR score of 12/24 (50%), reflecting moderate progress in metadata structuring and persistent identification while highlighting areas for improvement (Fig. 3).

The Findable (F) dimension received a score of 3.5/7 (50%), indicating that the data are uniquely identified and persistently referenced via PURL (<http://purl.org/raras>), but still require additional metadata elements for optimized searchability. The Accessible (A) dimension achieved 1.5/3 (50%), as metadata is retrievable via standardized protocols (HTTPS) but lacks explicit machine-readable access conditions. Interoperability (I) scored 3/4 (75%), demonstrating strong alignment with structured semantic vocabularies, such as Schema.org, Dublin Core,

Table 2 FAIR principles satisfied in the RARAS RDF schema and data

FAIR principles	RDF Schema	Compliant data	Tools/Services used
F1 Unique identifier	URLs for data elements and DOIs for external terminologies and datasets	URLs for data instances in persistent repositories	DOI Registration Services (via LattesData) Persistent URLs (via purl.org)
F2 Rich metadata	Metadata for data elements	Administrative process and descriptive metadata (described in the DMP and available in the RARAS Portal)	FairSharing ARGOS for DMP
F3 Linking data and metadata	RARAS RDF properties	Dataset linked to its metadata via unique identifiers	RDFa/Microdata
F4 Metadata searchable	Schema dereferenceable on the web	Searchable metadata fields are indexed in a repository and embedded in web pages.	RDFa/Microdata
A1 Open communication and authentication protocols	HTTPS protocol	Datasets accessible via the HTTPS protocol	APIs Typed Links HTTP Content negotiation
A2 Long-term availability	Available on the website, data repositories, and GitHub	Persistent data archived with version control	LattesData (a DataCite instance) Zenodo Figshare
I1 Knowledge representation	RDF	RDF	RDFa/Microdata D2Rq Server
I2 Controlled vocabulary	Meaning binding to ICD-10, Orpha, OMIM, and HPO codes	Use of standard vocabularies for metadata terms	REDCap, MeSH, BioPortal, Orphanet, and Monarch websites
I3 Linked data	RARAS and standard RDFS/OWL properties (i.e., rdfs: domain, rdfs: range, owl: equivalentClass, owl: Restriction)	RARAS RDF properties	Protégé D2Rq Server
R1.1 Data usage licence	Apache License, Version 2.0	Individual datasets may have their own license.	-
R1.2 Detailed provenance	Concepts and properties to represent administrative metadata	Tracking of data generation and transformations. h Administrative and descriptive metadata.	-
R1.3 Community standards	DataCite, Schema.org, Dublin Core, DCAT	Processed data available through widely accepted file formats and standards (e.g., CSV)	-

Evaluated Resource:

Atlas Brasileiro Online de Doenças Raras

FAIR level: moderate

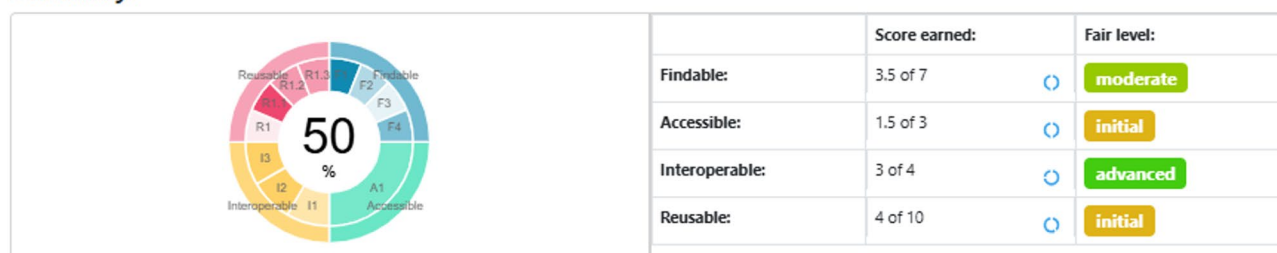
Resource PID/URL: <https://raras.org.br/>

DataCite support: enabled

Metric Version: metrics_v0.5

Metric Specification: <https://doi.org/10.5281/zenodo.6461229>

Software version: 3.4.0

Summary:**Fig. 3** F-UJI FAIR Data Assessment for RARAS FDO

and DCAT. However, the Reusable (R) dimension had the lowest score (4/10, 40%), mainly due to incomplete metadata descriptions and the absence of data format specifications. The full F-UJI FAIR assessment report is available in the Supplementary Material.

Regarding the governance model evaluation, the RE-AIM framework provides a valuable perspective for interpreting the results of this FAIR assessment. The Reach dimension is fulfilled primarily through RARAS, which ensures persistent access and unique identifiers, thereby supporting the broader accessibility of their findings and productions. Standardized metadata vocabularies support adoption and facilitate integration with other RD registries. However, the incomplete metadata structure presents a barrier to full adoption in automated data-sharing frameworks.

Regarding Implementation, the interoperability of RARAS FDO is well-established, demonstrating structured metadata representation through RDFa/Microdata using Schema.org, Dublin Core Terms, the DCAT and DataCite ontology. This confirms the system follows best practices for linked data management, enabling semantic integration with external datasets. While a SPARQL endpoint and RDF-based metadata extraction are available, access is not yet fully automated or integrated with external client applications. This partial implementation falls short of achieving complete alignment with FAIR standards, particularly in terms of programmatic accessibility [28]. Finally, Maintenance remains a key challenge, as the absence of automated metadata validation and registry updates poses a risk to the long-term sustainability of FAIR compliance [29].

Discussion

Data governance is the cornerstone of a federated data network's effectiveness. Successful scientific collaboration requires a booming governance model, where each database owner retains complete control over the use and access to patient-level data, ensuring compliance with ethical and legal standards [37]. This study has demonstrated that implementing a governance model is crucial for ensuring data integrity, privacy, and interoperability while promoting collaboration among research centers and healthcare institutions [38]. Integrating the FAIR principles into the RARAS network has showcased the potential to standardize data collection, enhance interoperability, and promote efficient data sharing, ultimately supporting advancements in scientific collaboration and clinical outcomes [11, 14].

Adopting the FAIRification process within the RARAS network highlights key advancements in digital health governance. The network successfully harmonized disparate data sources by establishing a comprehensive governance framework, transforming

them into FAIR-compliant datasets [25]. This process has enabled significant strides in data integration, improved analytical capabilities, and enhanced support for clinical decision-making [26]. The incorporation of semantic technologies, such as RDF and OWL, further facilitated data interoperability, allowing for more precise knowledge representation and enabling scalable research applications [39].

A key strength identified was the use of persistent identifiers and semantic interoperability mechanisms, ensuring data consistency across multiple systems. Applying the RE-AIM evaluation to the model, following the FAIR assessment, also provides a structured perspective on the RARAS network's impact and challenges [21, 29]. However, the evaluation revealed missing metadata elements, such as "object type," which impacts findability and reusability [40]. Moreover, while the system provides structured metadata, it does not yet support direct dataset access via machine-readable data links, which limits automated retrieval and integration into more extensive RD research infrastructures [11, 31].

Despite these successes, several challenges remain. A significant barrier to the FAIRification process was the limited digital literacy and technical capacity of stakeholders, particularly in clinical settings [32, 40]. Efforts to address this included targeted training sessions and iterative workshops, but ongoing education remains essential to sustain and expand the adoption of FAIR principles [32]. Data privacy and ethical concerns also posed challenges, requiring careful attention to data governance policies and consent management frameworks [14]. These challenges were compounded by the variability in data-sharing practices across regional centers, which hindered the uniform application of FAIR principles. Furthermore, the reliance on resource-intensive technologies and manual validation processes underscored the need for scalable and automated solutions [41].

Our first step was to understand the educational technology and resources needed to pave the way for RD Brazilian research [32] to meet the data quality culture needs, such as FAIR training for researchers, clinicians, and various organizational stakeholders [34]. Following this, we develop and implement a data governance policy that specifies how to extract and apply the data, as approved by the patient in the informed consent [33]. The delayed data governance may delay the FAIRification process because it needs to be clear what data will be available and in what form to plan the FAIRification [39]. Additionally, specifying data governance in the resource's metadata when an algorithm visits the data to use is also crucial [25, 40].

Coding errors and poor documentation are the primary reasons for data non-validation in information systems [14]. While post-submission curation can improve

data quality retrospectively, implementing practical data management solutions early in project design is essential [22]. These requirements aim to enhance clinical decision-making and are summarized by Kodra et al. in their perspective on 'quality informatics' using health data collected from information systems [6].

Due to the geographical dispersion and small number of patients, the transparency and reproducibility of real-world evidence in RD research are challenging. The federated data network approach offers a viable solution to these challenges [39]. Federated data networks enable the harmonization of data from multiple sources, thereby enhancing the robustness and credibility of observational research. Evolving the RARAS network to a federated model could potentially facilitate the integration of clinical, genomic, and other data types, promoting a more comprehensive and detailed analysis of RD data [37].

Compared to the RARAS network, international efforts such as the ERNs have faced similar challenges in integrating data across multiple centers and countries. One of the primary bottlenecks in the FAIRification of RD data has been the lack of standardized metadata, which hinders data interoperability across platforms and research institutions [9]. Similar challenges have been encountered by RARAS, particularly in ensuring the consistent application of metadata standards across diverse centers [12].

The ERNs have implemented solutions such as using standard data models and establishing centralized repositories to promote data sharing. RARAS follows a similar path by integrating data from reference centers and ensuring adherence to FAIR principles [24]. However, like ERNs, RARAS faces the ongoing challenge of maintaining data quality and ensuring the long-term sustainability of its data infrastructure [9, 34].

Finally, data harmonization and the standardization of terminologies are essential for the effectiveness of federated data networks. van Baalen et al. (2024) emphasize that data harmonization through a standard data model is fundamental for ensuring the consistency and comparability of data across different sources [37]. For the RARAS network, adopting a standardized MDS facilitated data integration from diverse sources [33], improved data quality [34], and enabled more accurate and practical analysis, significantly contributing to the advancement of RD research in Brazil.

The research underscores the importance of data quality and semantic mapping. Data from diverse institutions undergo a harmonization process, mapped to agreed-upon terminologies, facilitating seamless queries within a federated network [37]. It can also lead to the construction of a knowledge base, which should be capable of interacting with other data systems. The secondary use of health data is crucial for enhancing healthcare quality

and improving the efficacy of research [38]. Aligned with appropriate digital solutions, it can enable testing, validation, and comparison of artificial intelligence solutions and big data analytics across various parameters and settings, thus continuing patient care [41].

Our study has some limitations: website service quality is not a fully defined construct, and there remains uncertainty in explaining and interpreting its meaning, primarily due to the variety of site types [24]. As such, there is no established conceptual model for developing and evaluating website service quality. Additionally, while the RARAS platform uses anonymized data, patient rights related to data access and deletion are governed by Brazil's General Data Protection Law [1]. Given the decentralized nature of the public health system, these rights are implemented at the institutional level, which may lead to variations and delays in requests across centers [12, 38].

Furthermore, some tools presented specific limitations. While REDCap effectively supports data collection, it requires extensive manual interventions for data interoperability and lacks robust automated validation, which demands significant IT resources [11]. FAIRsharing and ARGOS facilitate metadata management but require customization to accommodate RARAS's large-scale, federated nature [15].

Their lack of built-in mechanisms for longitudinal metadata updates poses challenges in maintaining FAIR compliance [40]. While registration in FAIRsharing is an essential step toward enhancing the findability and interoperability of datasets, it primarily focuses on metadata-related aspects of FAIRness [25]. It does not, by itself, ensure overall FAIR compliance, especially in dimensions such as accessibility, reuse conditions, or semantic interoperability. These require additional governance, technical infrastructure, and validation processes [11].

Another limitation is the reliance on external digital resources, such as Orphanet, whose periodic updates and partial content availability in Portuguese complicate the standardization of RD classification. Brazil's low English literacy rates, combined with the limited experience of data collectors, exacerbate this issue [32]. Additionally, the absence of a unified RD classification system hinders interoperability, requiring additional efforts to align datasets [42].

To address some of these challenges, the RARAS web portal functions as a central digital asset for the Brazilian RD ecosystem. It publishes epidemiological data from health services and provides tools and services tailored to RD stakeholders. In this sense, our model was developed as a composite of digital objects available on the portal, all of which are designed to be FAIR and assessable [33]. These components enable the monitoring of FDO usage

across various users and settings. Additionally, feedback surveys available on the portal offer qualitative insights into factors that facilitate or hinder FDO adoption [13].

As our evaluation analyses cover only a specific period, it is desirable to understand the portal's long-term scalability and maintenance [30]. Addressing these tool-specific limitations in future iterations could enhance the robustness, efficiency, and scalability of the presented digital governance model. Future efforts should focus on improving the scalability and sustainability of the FAIRification process [25]. This includes automating data validation and annotation tasks, developing user-friendly tools for stakeholders with varying technical expertise, and strengthening partnerships with global data-sharing initiatives [41]. Additionally, integrating federated learning models could address privacy concerns while enabling advanced analytics across distributed datasets [37].

To address these gaps and optimize the RE-AIM dimensions evaluation, targeted improvements should focus on (1) expanding metadata completeness by including missing descriptors such as “object type”; (2) enhancing accessibility through persistent dataset links; (3) ensuring broader adoption by registering metadata in global repositories; (4) increasing implementation efficiency by integrating machine-readable access mechanisms; and (5) securing long-term maintenance through automation of metadata validation and updates. These measures will contribute to a more robust and FAIR-aligned governance framework for RD research in Brazil.

Conclusion

The research highlights the need for standardized practices and enhanced collaborative efforts to improve data quality in Brazilian RD research. Despite the focus on RD research, the presentation of practical implications is extensive and notably pertinent to health policy, clinical practice, and the design of health information systems [20]. The described FDO and process, including the FAIRification workflow and a scalable governance framework, directly apply to enhancing health data management and sharing in Brazil.

Our findings offer system developers clear guidelines for creating efficient and user-friendly health information systems. These findings can be extended beyond RD research, offering valuable insights into broader applications of digital health governance. The developed framework provides a replicable model for enhancing data quality, promoting interoperability, and facilitating collaborative research in other healthcare domains. Policy-makers and practitioners can leverage these insights to design and implement data governance strategies that align with international standards while addressing local needs [6].

Our findings address the research question by implementing a digital health governance model grounded in FAIR principles, thereby significantly enhancing data quality. This study contributed to laying the foundations for the long-term maintenance and availability of Brazil's first national registry of RDs by advancing collaborative efforts, data standardization, and FAIR-aligned governance. By applying standardized data collection, ensuring interoperability through semantic technologies, and fostering collaboration among key stakeholders, we have established a sustainable and scalable infrastructure that enhances data quality, accessibility, and the impact of collaborative research on the future of RD research in Brazil. It also contributes to Sustainable Development Goal 3 (Good Health & Well-Being), enhancing data-driven decision-making, supporting evidence-based health policies, and fostering global collaboration in RD research.

Finally, the interdisciplinary nature of the results underscores its significant strength. By integrating insights from data science, public health, and IT governance, this study opens a broader perspective on the challenges in digital health. This collaborative approach is essential for addressing complex public health issues, as interdisciplinary perspectives can yield more innovative and effective solutions. Integrating diverse methodologies and perspectives in the dissertation underscores the importance of multidisciplinary collaboration in advancing digital health.

Abbreviations

API	Application Programming Interface
CSV	Comma-Separated Values
CNPq	Conselho Nacional de Desenvolvimento Científico e Tecnológico
DMP	Data Management Plan
DOI	Digital Object Identifier
DCAT	Data Catalog Vocabulary
EDC	Electronic Data Capture
ERN	European Reference Network
FAIR	Findable, Accessible, Interoperable, and Reusable
FDO	FAIR Digital Object
GPL	General Public License
HPO	Human Phenotype Ontology
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
ICD-10	International Classification of Diseases, 10th Revision
IRI	Internationalized Resource Identifier
JSON	JavaScript Object Notation
JSON-LD	JSON for Linked Data
LattesData	Brazilian Open-Access Repository for Research Data
MDS	Minimum Data Set
MeSH	Medical Subject Headings
NIH	National Institutes of Health
OWL	Web Ontology Language
PROM	Patient-Reported Outcome Measure
RARAS	Rede Brasileira de Doenças Raras (Brazilian Rare Disease Network)
RD	Rare Diseases
RE-AIM	Reach, Effectiveness, Adoption, Implementation, and Maintenance Framework
REDCap	Research Electronic Data Capture
RDF	Resource Description Framework

WHO World Health Organization
Zenodo Open Science Data Repository

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s44247-025-00200-5>.

Supplementary Material 1

Supplementary Material 2

Supplementary Material 3

Supplementary Material 4

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Author contributions

F.B., V.L., R.R., T.F., and D.A. conceptualized the study, designed the methodology, and supervised the project. B.O., C.L., A.V., D.Y., G.S., Y.A., M.S., and G.A. participated in data collection, curation, and preliminary analysis. M.A., J.C.M., V.C., and J.B. contributed to developing the digital infrastructure and FAIR modeling. F.B., V.L., B.O., C.L., V.F., and T.F. conducted the data analysis, interpretation, and validation. All authors contributed to drafting and revising the manuscript critically for important intellectual content and approved the final version.

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Data availability

Availability of data and materials statement The data described in this article can be freely and openly accessed at LattesData Dataverse: <https://doi.org/10.57810/lattesdata/XEL530>. The analysis and tools are publicly available on GitHub: <https://github.com/RARAS-BR>. This study's metadata and associated standards are registered and accessible via the FAIRsharing record (<https://fairsharing.org/10.25504/FAIRsharing.d7b6c8>) and the DMP at the ARGOS record, which has the DOI 10.5281/zenodo.13643431 and is available at <https://argos.openaire.eu/explore-plans/overview/public/8d4a6851-f96e-4a29-ad79-37e8bf30d17a>. The intellectual property of the developed materials is registered with the Brazilian National Institute of Industrial Property (INPI) under the process number BR5120220032899.

Declarations

Ethical approval and consent to participate

It is not applicable. This study did not involve human participants or tissue. Ethics approval and consent to participate are not applicable, as this study was based solely on publicly available, anonymized datasets for analysis. This process removes direct identifiers, making it impossible to trace data back to individual patients without additional information. All data used in the study were pseudonymized before inclusion in the RARAS database. It did not involve recruiting new human participants or collecting data. However, the primary study that originated this research was approved by the Research Ethics Committee (REC) of the Hospital de Clínicas de Porto Alegre, Brazil (approval number: 33970820.0.1001.5327).

Consent for publication

Not applicable. This manuscript contains no person's data (including individual details, images, or videos).

Competing interests

The authors declare no competing interests.

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